Power Market ML

A learning guide to the market concepts, data work, statistics and machine learning behind the project, and how each result was tested.

Companion to github.com/krithik12345/power-market-ml

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0 How to use this guide

This guide teaches the ideas behind the **power-market-ml** repository: how wholesale electricity markets work, how to turn public EIA data into a clean modeling dataset, and how to test market claims and forecasts without fooling yourself. Every concept is tied to a specific file and a real result from the repo, so you can read an idea here and then see it running in code.

Who it is for

- Students and engineers who know some Python and want to understand how power markets work.
- People who already know power markets and want to see how statistics and machine learning are used to test trading ideas.
- Anyone reading the repo who wants to understand *why* each design choice was made, not just what the code does.

What you should know first

- **Python and pandas:** reading a DataFrame, filtering, merging, and groupby. You don't need to know any modeling libraries.
- **Basic statistics:** mean, median, percentiles, and what a linear regression line is. Everything beyond that is explained here.
- **No power-market background needed.** Part 2 starts from zero.

How to read it

Parts 1 to 6 build up the concepts in order: markets, data, statistics, models, evaluation. Part 7 applies all of them to the four studies on the EIA hubs, Part 8 extends the project to Texas with hourly ERCOT data, Part 9 moves beyond EIA to the ISOs' own data, and Part 10 covers what went wrong along the way. If you only have an hour, read Part 1, the 'key idea' notes, and Parts 7 to 9. Each part ends with a **Check yourself** list. If you can explain every item out loud without looking, you've got that part.

Run it yourself

Reading is good; changing the code and predicting what happens is better. Setup takes about ten minutes:

```
git clone https://github.com/krithik12345/power-market-ml
cd power-market-ml
python -m venv .venv
.venv/Scripts/pip install -r requirements.txt   # macOS/Linux: .venv/bin/pip
copy .env.example .env                          # macOS/Linux: cp
                                                # then paste a free key from
                                                # eia.gov/opendata/register.php
python scripts/fetch_data.py                    # downloads and caches all data
python scripts/thesis_03_pjm_convexity.py       # reproduces one study
```

1 The big picture

The project takes free public data from the U.S. Energy Information Administration (EIA) and uses it to do two different jobs. Keep these two jobs separate in your head; most of the design decisions follow from the difference.

	Job A: Thesis testing	Job B: Forecasting
Question	*Why* do prices behave the way they do? Is a market claim true?	*What* will tomorrow's price be?
Example	Does more solar push SP15 prices down relative to gas?	Predict tomorrow's SP15 on-peak price this morning
Allowed information	Anything about the delivery day, including what actually happened	Only what a trader knows on the trade date
Main tools	Regression with controls, standard errors, interaction terms, quantile regression	Persistence, ridge, LightGBM, walk-forward backtests
Success looks like	A coefficient with the right sign, a tight confidence interval, and robustness checks it survives	Lower error than a simple baseline on years the model never saw
Scripts	thesis_01, 01b, 02, 03	forecast_benchmark.py (and part of T02 and T03)

The pipeline in five steps

- **Download** daily power prices for five trading hubs, daily Henry Hub gas prices, and daily grid data (demand, demand forecast, generation by fuel) for the grid operator behind each hub. Cache everything locally.
- **Align** them into one table per hub, one row per delivery day.
- **Engineer features**: the implied heat rate, fuel shares, net load, calendar terms, and lagged values that a trader would actually know.
- **Test theses** with regressions designed to isolate one effect at a time, then try to break each result with robustness checks.
- **Forecast** with several models, graded only on years they never trained on, against a hard baseline.

> THE KEY IDEA
>
> The single most important modeling choice: the project studies the **implied market heat rate** (power price divided by gas price) instead of the raw power price. Gas prices swing a lot and drag power prices with them, which hides everything else. Dividing by gas removes most of that, so what's left is what the grid itself is doing: solar, wind, hydro and load.

What was found

Study	Result	Status
T01 Solar at SP15	Spring median heat rate fell from 10.3 (2019) to 2.1 (2026) as spring solar share rose from 15% to 25%. +10 points of solar is about -7.9 heat-rate units, pooled.	Supported
T01b Batteries	A bigger battery fleet does not measurably weaken solar's effect on the daily on-peak price. The batteries were found hidden inside EIA's 'Other' category.	Not supported
T02 Mid-C hydro	+1 GW of Northwest hydro above normal moves Mid-C about $6/MWh lower vs SP15. Mid-C flipped from a discount to a premium after 2022.	Supported
T03 PJM spikes	Heat rate is convex in load; the top 10% of load days carry 39% of summer upside. Spike days are predictable a day ahead (AUC 0.93 vs 0.84 baseline).	Supported
F01 Forecasts	Models beat 'tomorrow = today' at all five hubs: 20% better at PJM West, 2% at Mid-C.	Done
E1 ERCOT peak hour	The most expensive hour moved from 5pm (demand peak) to 8pm (net-load peak). Days priced near the demand peak fell from 74% to 13%.	Supported
E2 ERCOT batteries	The evening ramp more than doubled but its price per GW fell from 2.2 (2023) to 0.1 (2026) as batteries grew to about 9 GW.	Supported, with a caution
D1 Duck curve	Storage now removes 55% of CAISO's evening ramp and 39% of ERCOT's (0% in 2019). CAISO's storage-adjusted ramp is shrinking.	Parts 1-2 supported; part 3 not (biased test)
E3 ERCOT real time	Each GW of batteries cuts evening real-time spikes above day-ahead. The day-ahead 'insurance premium' fell from $17.49 to $0.72/MWh.	Supported
E4 ERCOT reserves	Reserve prices fell about 26-30% per GW of batteries; RRS went from $17 (2021) to $0.71/MW-h (2026).	Supported

2 Power market concepts

2.1 Hubs, ISOs and balancing authorities

Electricity is traded at **hubs**: named pricing points that stand in for a region. A hub price is usually an average of locational prices at many nodes. The five hubs in this project:

Hub	Region	Grid operator (EIA-930 code)	What drives it
SP15	Southern California	CAISO (CISO)	Solar, gas, imports
Mid-C	Pacific Northwest (Mid-Columbia river)	Bonneville Power (BPAT)	Hydro, wind
Palo Verde	Arizona	APS + SRP (AZPS, SRP)	Solar, gas, nuclear
PJM West	Mid-Atlantic	PJM (PJM)	Gas, coal, load
Mass Hub	New England	ISO-NE (ISNE)	Gas (pipeline-constrained in winter)

An **ISO/RTO** (Independent System Operator / Regional Transmission Organization) runs the grid and the wholesale market in its region. A **balancing authority (BA)** is any entity responsible for keeping supply and demand balanced in an area; every ISO is a BA, but some BAs (like BPA or APS) are utilities. EIA's Form 930 reports data by BA, which is why the project maps each hub to one or more BAs.

2.2 Day-ahead, real-time, and the on-peak block

- **Day-ahead (DA):** power for tomorrow, traded today. Most hedging happens here.
- **Real-time (RT):** the price that actually clears in the moment. Spikier, because surprises (a plant trips, a heat wave arrives early) show up here.
- **On-peak block:** a standard product covering hour-ending 7 through 22 (7am-10pm) on weekdays. The EIA/ICE files give one price per hub per day for this block. That means every price in the project is an *average over 16 hours*, which matters a lot in T01b.
- **Trade date vs delivery date:** a day-ahead trade on Monday is for delivery on Tuesday. The project keys every row by delivery date, but uses the trade date to decide what information was available. Friday trades usually cover Monday delivery; multi-day weekend packages were dropped.

2.3 The supply stack (merit order)

Power plants are dispatched cheapest-first. Picture all plants lined up by their cost to produce one more MWh: renewables and nuclear (near zero), then coal and efficient gas, then older gas plants, then peakers that run only a few hours a year. Demand is a vertical line on this chart; the **last plant needed to meet demand sets the price** for everyone.

- The stack is flat for a long stretch and then turns sharply upward at the end. This is a **hockey stick**, or **convex** shape.

- Consequence: a 5 GW rise in load on a mild day barely moves the price, but the same 5 GW on the hottest day of the year can double it. This is exactly what T03 measures.
- Solar and wind sit at the cheap end. Adding them shifts the whole stack right, so a given load is met further down the curve. This is the mechanism behind T01.

2.4 Heat rate and spark spread

Physical heat rate is how much fuel a plant burns per MWh it produces, in MMBtu/MWh. Lower is more efficient. A modern combined-cycle gas plant is about 6.5-7.5; an old peaker can be 10-14.

Implied market heat rate (IHR) turns this around: it asks what heat rate the *market price* implies.

```
IHR = power price ($/MWh) / gas price ($/MMBtu)        units: MMBtu/MWh

Example: power $45/MWh, gas $3.00/MMBtu  ->  IHR = 45 / 3 = 15
```

An IHR of 15 says the market is paying as if a fairly inefficient plant is setting the price, so the system is tight. An IHR of 3 says power is cheap relative to its fuel, so something other than gas (solar, hydro) is setting the price.

Spark spread is the gross margin of a gas plant: power price minus fuel cost.

```
spark spread = power price - (plant heat rate x gas price)

Example: $45 - (7 x $3.00) = $45 - $21 = $24/MWh margin for a 7-heat-rate plant
```

> THE KEY IDEA
> A trader who is 'short heat rate' profits if power gets cheaper *relative to gas*, whatever gas itself does. That's why heat rate is the natural unit for the theses: it's the unit a power trader actually takes a view on.

2.5 Net load, the duck curve, and solar cannibalization

Net load = demand minus wind minus solar. It's the load the rest of the fleet (gas, hydro, imports) must cover, and it's usually a better price driver than raw demand. In California, net load collapses at midday when solar peaks and then shoots up at sunset. Plotted over a day, this looks like a duck (the **duck curve**).

Cannibalization: each new solar plant produces at the same hours as every existing one, pushing midday prices down and reducing the value of all solar. T01 measures this at the daily on-peak level.

2.6 Basis and spreads

Basis is the price difference between two locations (or a location and a benchmark). The Mid-C minus SP15 spread in T02 is a basis. Trading a spread (long one hub, short another) removes the exposure the two share, such as western gas prices, and leaves a bet on what differs between them, such as Northwest hydro.

Gas basis matters too. Henry Hub (Louisiana) is the national benchmark, but western power plants buy gas at regional hubs (SoCal Citygate, Sumas, PG&E Citygate). In winter 2022-23, western gas spiked to

several times Henry Hub. Since the project divides by Henry Hub, western IHRs in those months look absurdly high. This is why the analyses **winsorize** outliers and flag 2022-23 everywhere.

2.7 Storage and intraday shape

Batteries charge when power is cheap (midday solar) and discharge when it is expensive (the evening ramp). This narrows the gap between midday and evening prices: the **intraday shape**. Because the on-peak block contains both midday and evening hours, a battery moving energy from noon to 8pm mostly reshuffles value *inside* the block. That's the key insight from T01b.

2.8 Hydro

Northwest hydro output depends on snowpack and spring runoff. It peaks around May-June. What matters for prices is not the season itself (the market already expects spring runoff) but the **anomaly**: how much more or less water there is than normal for that month.

> CHECK YOURSELF: CAN YOU EXPLAIN
>
> – Why a heat rate of 15 means a tight market, and how to compute it from a price and a gas price.
> – Why the supply stack makes prices convex in load.
> – What net load is and why it predicts price better than demand.
> – Why a spread trade isolates one driver.
> – Why dividing by Henry Hub distorted the western numbers in 2022-23.

3 Data engineering

3.1 The three sources

Source	What one row is	How it's fetched	Code
EIA wholesale (ICE) power	One hub, one trade date, one delivery period: weighted average price, high, low, volume, trades	Yearly Excel files, 2017-2026, downloaded once and cached	ice_power_prices( )
Henry Hub spot (RNGWHHD)	One trade date, one $/MMBtu price	One Excel file covering 1997 to today, refreshed daily	henry_hub()
EIA-930 grid data	One BA, one day (or hour), one series: demand, demand forecast, or generation by fuel	EIA API v2 with a free key, paginated, cached as Parquet	eia930_daily(), eia930_hourly()
ERCOT day-ahead prices (Part 8)	One ERCOT hub, one delivery date, one hour: price in $/MWh	Yearly zip files from ERCOT's public reports site, no key	ercot.dam_hub_pr ices()

Findings during data collection that shaped the project: ERCOT North stopped appearing in the ICE files after 2018, so Texas prices come from ERCOT itself (Part 8). NP15 and Indiana trade too thinly for daily modeling. The regional gas files EIA used to post are no longer available, so Henry Hub is the only gas benchmark. CAISO does not report batteries as their own fuel type.

3.2 Working with an API

- **Routes and facets:** EIA API v2 is organized by routes (e.g. `electricity/rto/daily-fuel-type-data`). Facets are filters: `respondent=CISO`, `fueltype=SUN`, `timezone=Pacific`.
- **Pagination:** the API returns at most 5,000 rows per request. The client loops, increasing `offset` by 5,000 until it has `total` rows.
- **Rate limits and retries:** HTTP 429 means 'too many requests'. The client waits and retries with **exponential backoff** (5s, 10s, 20s, 40s). EIA's public DEMO_KEY is too rate-limited to download everything, so you need your own free key (Part 0).
- **Secrets:** the key lives in `.env`, loaded with python-dotenv, and `.gitignore` keeps it out of git.
- **Caching:** every download is saved (Excel in data/raw, API results as Parquet in data/cache). Re-runs are instant, offline, and don't burn API quota. Parquet is a compressed columnar format that loads much faster than CSV and keeps data types.

3.3 Lining the data up

This is where most real-world bugs live. Four alignment problems were solved:

- **Keys:** power rows are keyed by delivery date; gas rows by trade date. Gas is joined on the power trade date because both are next-day products traded the same morning.
- **Missing days:** Henry Hub doesn't print on some holidays. `pd.merge_asof(direction='backward')` takes the most recent earlier gas price (up to 5 days back) instead of leaving a gap.

- **Time zones:** hourly EIA-930 data is in UTC. To compute a California 'day', the battery code converts to US/Pacific before grouping by date; otherwise the evening discharge would land on the wrong day.
- **Units:** daily grid totals are MWh per day. Dividing by 24 and by 1000 gives **average GW**, a number people can picture (CAISO averages roughly 25-35 GW).

3.4 Feature engineering

Feature	Formula	Why
ihr	price / gas	Gas-neutral price (section 2.4)
solar_share, wind_share, hydro_share	generation by fuel / demand	Scale-free: comparable across years as demand grows
net_load_gw	(demand - solar - wind) / 24 / 1000	What the rest of the fleet must cover
doy_sin, doy_cos	sin and cos of 2 pi x day-of-year / 365.25	Encodes season smoothly: Dec 31 and Jan 1 end up next to each other, which a 1-365 number can't do
dow, month	calendar	Weekday and seasonal patterns; used as categories (one-hot or fixed effects)
trend_yrs	years since 2019-01-01	Lets models capture slow structural change
price_lag, ihr_lag	previous delivery day's value	The persistence signal
price_lag_5d_mean	rolling 5-day mean of past prices	Smoother recent level
demand_fcst_delta	tomorrow's demand forecast - yesterday's actual demand	Expected change in load: the most useful signal in T03
gas_chg	gas today - gas on previous price day	Fuel-cost shock
*_lag (fuel shares)	values from the day before the trade date	What was actually knowable at trade time

3.5 Look-ahead bias: 'explain' vs 'known' features

A forecast that uses information from the future looks brilliant in a backtest and fails in real life. This is called **look-ahead bias** or **leakage**. The project prevents it by splitting features into two lists in `features.py`:

- **EXPLAIN**: what actually happened on the delivery day (actual solar, actual demand). Used only for thesis testing, where the question is 'why'.
- **KNOWN**: what a trader has on the trade-date morning: the demand *forecast* for tomorrow, yesterday's realized mix, today's gas, the last cleared price. Used for every forecast.

CAUTION

Even 'known' features rest on an assumption: that EIA's day-ahead demand forecast is published before the morning trading window. EIA doesn't document the exact time. A live system should use the ISO's own forecast, which is published well before trading.

3.6 Recovering hidden batteries (T01b)

CAISO doesn't report a BAT fuel type to EIA-930. A query for every storage code returned nothing, but the 'Other' (OTH) category had a slightly *negative* daily total. That's a clue, since storage loses energy on each charge-discharge cycle. Pulling OTH **hourly** and averaging by hour of day showed it:

Hour (Pacific)	2019 average MW	2026 average MW	Meaning
12 (noon)	-3	-7,163	Charging on midday solar
20 (8pm)	+5	+7,595	Discharging into the evening ramp

The daily battery features are the sum of positive hours (discharge) and of negative hours (charge). The fleet-size proxy is the **trailing 90-day 95th percentile** of daily discharge, shifted by a day. Section 4.9 explains why it's built this way.

> CHECK YOURSELF: CAN YOU EXPLAIN
>
> - Why the API client paginates and backs off, and why results are cached.
> - What merge_asof does and why gas is joined on the trade date.
> - Why sin/cos encode season better than day number.
> - The difference between EXPLAIN and KNOWN features, and what leakage would look like.
> - How the batteries were found in data that doesn't label them.

4 Statistics for testing theses

4.1 Ordinary least squares (OLS) regression

OLS fits a straight-line relationship by minimizing the sum of squared errors. The T01 model:

```
ihr = b0 + b1*solar_pp + b2*wind_pp + b3*hydro_pp + b4*demand_gw
      + month effects + weekday effects + error
```

Reading a coefficient: b1 is the expected change in IHR for a 1-point increase in solar share, *holding everything else in the model constant*. T01 got b1 = -0.785, reported per 10 points as -7.85. Shares were converted to percentage points (x100) precisely so the coefficient reads naturally. **R^2** is the fraction of variation explained (0.41 in T01): useful for comparing models on the same data, but not a measure of whether a coefficient is right.

4.2 Controls and fixed effects

Solar is high in spring, and spring also has mild load and high hydro. Without controls, the solar coefficient would absorb all of that. Adding wind, hydro and demand as **controls** means the solar effect is estimated after accounting for them.

`C(month)` and `C(dow)` are **fixed effects**: a separate intercept for each month and weekday. They absorb anything that is typical of 'April' or 'Monday'. The solar coefficient is then identified from days that had *more or less solar than usual for that month*, which is a much cleaner comparison.

4.3 Standard errors, confidence intervals, p-values

- The **standard error (s.e.)** measures how much a coefficient would bounce around if you had a different sample of days.
- A **95% confidence interval** is roughly coefficient ± 1.96 x s.e. T01: -7.85 with CI -9.39 to -6.31. Since the whole interval is below zero, the effect is clearly negative.
- The **p-value** is the probability of seeing an effect this large if the true effect were zero. Below 0.05 is the usual threshold for 'significant'. It is *not* the probability the thesis is true.

4.4 HAC (Newey-West) standard errors

Ordinary standard errors assume each day's error is independent. Market days are not independent: a heat wave or a gas squeeze lasts several days, so errors are **autocorrelated**. Ignoring that makes standard errors too small and results look more certain than they are.

HAC (heteroskedasticity- and autocorrelation-consistent) standard errors, also called Newey-West, correct for this. Every regression in the project uses `cov_type='HAC', maxlags=5`, allowing correlation across about a trading week.

4.5 Interaction terms

An interaction lets one variable's effect depend on another. Two were used:

- **solar x year** (T01): estimates a separate solar slope for each year, so you can see if the effect is changing over time.
- **solar x battery** (T01b): asks whether solar's effect depends on how big the battery fleet is.

```
ihr = b1*solar + b2*battery + b3*(solar x battery) + ...

marginal effect of solar = b1 + b3*battery
   b3 > 0  ->  more batteries make solar's (negative) effect smaller
```

4.6 The delta method

To put a confidence band around 'the effect of solar at battery level b', you need the standard error of a *combination* of coefficients. The delta method gives it from the coefficient covariance matrix V:

```
Var(b1 + b*b3) = V[1,1] + b^2 * V[3,3] + 2*b*V[1,3]
```

That's the blue shaded band in the T01b chart (`solar_effect()` in the script).

4.7 Pooled vs within-year variation

T01 found a pooled slope of -7.85 but per-year slopes between about 0 and -11. Both are right; they answer different questions. The pooled slope also picks up the **between-year** pattern (years with more solar had lower heat rates overall, partly for other reasons such as gas prices and new transmission). The per-year slopes only use **within-year** day-to-day variation, which is a cleaner estimate of solar's direct effect. When the two differ a lot, something else is changing over time.

4.8 Outliers and winsorizing

Winsorizing caps extreme values at a percentile instead of deleting them (T01: 1st and 99th; T03: 0.5th and 99.5th). One absurd day from the 2022-23 gas crisis can't then dominate a regression that minimizes *squared* errors, since squaring makes big misses count far more.

4.9 Confounding, reverse causality and collinearity

These are the three ways a regression can mislead you, and T01b ran into all three:

- **Confounding:** something else drives both variables. Battery fleet size grows steadily over time, as do transmission, exports and demand response. Any of them could be what the 'battery' term picks up. The fix was a competing **solar x time trend** term to see if batteries still mattered after accounting for time.
- **Reverse causality:** batteries discharge *more* on high-price days, so same-day discharge is partly caused by price. Using it as a regressor would bias the result. The fix was to measure fleet *capability* (trailing 90-day 95th percentile, shifted one day), which today's price can't affect.
- **Collinearity:** when two regressors move almost together (battery fleet and time), the model can't tell them apart, and standard errors blow up. That's why the '+ time trend' row in T01b has a much larger s.e. (0.095 vs 0.023).

4.10 Robustness checks

A result you believe should survive reasonable changes to how it's estimated. T01b ran three versions: main model, plus a time trend, and excluding 2022-23. The verdict logic in the script requires the effect to hold up across these before calling it 'supported'.

4.11 Quantile regression (T03)

OLS models the **average**. Quantile regression models a chosen **percentile**: the median (P50), the bad days (P90), the quiet days (P10). It minimizes a tilted absolute error called the pinball loss:

```
pinball loss for quantile q:   q * (y - pred)      if y > pred   (under-predicted)
                              (1-q) * (pred - y)   if y < pred   (over-predicted)

For q = 0.9, under-predicting costs 9x more than over-predicting,
so the fitted line sits where about 90% of points fall below it.
```

T03 fits `ihr ~ load + load²` at P10, P50 and P90. The **squared term** allows curvature, and the slope at any load is the derivative b1 + 2 x b2 x load. If the P90 line steepens faster than the P50 as load rises, the risk is concentrated in the tail: **convexity**.

CAUTION

A quadratic must curve the same way everywhere, so the P90 curve also bends up at the *low*-load end of the chart. That's a fitting artifact, not a real effect. Splines or binned estimates would avoid it.

4.12 Anomalies vs climatology (T02)

Climatology is the normal value for a time of year (here, average hydro output for each calendar month). The **anomaly** is actual minus normal. Regressing on the anomaly instead of the raw level avoids confusing 'it's spring' with 'it's a wet spring'. In the forecast test, climatology is rebuilt from *past years only* for each test year, so the 'normal' never includes the future.

CHECK YOURSELF: CAN YOU EXPLAIN

- How to read a coefficient in its units, and what 'holding constant' means.
- What month fixed effects do.
- Why HAC standard errors are needed for daily market data.
- How an interaction term changes the interpretation, and how the delta method builds the CI.
- Why pooled and per-year slopes differ.
- Confounding vs reverse causality vs collinearity, each with the T01b example.
- What quantile regression measures that OLS can't.

5 Machine learning models

5.1 Persistence: the baseline

Persistence predicts that tomorrow equals today. It sounds naive, but power prices are strongly autocorrelated, so it's very hard to beat. Any model that can't beat persistence has learned nothing useful. Every result in the project is reported against it.

5.2 Ridge regression

Ridge is linear regression with a penalty on large coefficients:

```
minimize   sum (y - Xb)^2   +   alpha * sum b^2
```

- The penalty (**regularization**) shrinks coefficients toward zero, which reduces overfitting when features are correlated, as these are.
- `RidgeCV` picks `alpha` from 20 values between 0.01 and 1000 by internal cross-validation.
- **Standardization** (`StandardScaler`) rescales each feature to mean 0, standard deviation 1. Without it, the penalty would punish features measured in large units differently from ones in small units.
- **One-hot encoding** turns month and weekday into 0/1 columns so the model doesn't treat December (12) as 'more' than January (1).
- A **Pipeline** plus **ColumnTransformer** bundles scaling, encoding and the model, so the scaler is fit only on training data. Fitting it on all data would be a subtle leak.

5.3 Target transformation: predicting heat rate, then multiplying by gas

`HeatRateRidge` learns to predict IHR, then multiplies the prediction by the known gas price to get $/MWh. Gas levels moved from about $2 to $9 during the sample; a linear model in dollars would need different coefficients in each gas regime, but heat rate is far more stable. This model won at PJM West and Mass Hub, where prices really are gas-linked.

5.4 Gradient-boosted trees (LightGBM)

A **decision tree** splits the data with yes/no questions (Is demand forecast above 110 GW? Is it July?) and predicts the average of each leaf. **Gradient boosting** builds hundreds of small trees in sequence; each one is fit to the errors the previous trees still make. LightGBM is a fast implementation.

Setting used	Value	What it controls
n_estimators	300-400	Number of trees
learning_rate	0.03	How much each tree contributes. Small + many trees = steadier
num_leaves	15	Tree complexity. Small = less overfitting
min_child_samples	20	Smallest allowed leaf; stops fitting to a handful of days
subsample / colsample_bytree	0.8 / 0.8	Each tree sees 80% of rows and features: adds randomness that reduces overfitting

Trees capture **nonlinearities** and **interactions** automatically (e.g., high load matters more in summer), which linear models need to be told about.

5.5 Why trees can't extrapolate, and the fix

THE KEY IDEA

A tree predicts the average of training examples in a leaf, so it **can never predict a value outside the range it was trained on**. A LightGBM model trained through 2021 had never seen 2022's gas-crisis prices, so it kept predicting 2021-sized prices. On price levels it lost badly to persistence (MAE 18.97 vs 11.04 at SP15).

The fix (`OnPersistence`): train the model on the **change** from the last price (y - price_lag), then add the last price back. Persistence becomes the floor, and the model only learns adjustments, which stay in a stable range in every regime. The same model then beat persistence in every test year (MAE 10.12). This is often called **residual modeling** or **modeling on a baseline**.

5.6 Loss functions

Loss	Penalizes	Used for
Squared error (L2)	Big misses heavily	Ridge, LightGBM on levels. Predicts the mean
Huber	Like L2 for small errors, like absolute error for large ones	LightGBM on change. Robust to spike days dominating the fit
Quantile (pinball)	Asymmetric (section 4.11)	P10 and P90 forecast bands
Log loss	Confident wrong probabilities	Logistic regression and LightGBM classifier (T03)

5.7 Classification for spike days (T03)

- **Logistic regression** is a linear model passed through the S-shaped logistic function, so it outputs a probability between 0 and 1.
- **LightGBM classifier** is the boosted-tree version, also outputting a probability.
- Both used the same known features. Logistic won slightly (AUC 0.93 vs 0.91). On a small, low-noise problem, a simple model often matches or beats a complex one.

5.8 Partial dependence

A partial dependence plot shows a model's average prediction as one feature changes while the others keep their actual values. T01 used it to check whether solar's effect is curved: the GBM's predicted IHR fell from 20.0 at 8% solar to 12.6 at 29%. It shows what the *model* learned, which is a tool for interpretation, not proof of cause.

CHECK YOURSELF: CAN YOU EXPLAIN

- Why persistence is a strong baseline for power prices.
- What regularization does and why features are standardized before ridge.
- How gradient boosting works in two sentences.
- Why a tree model can't forecast a price regime it hasn't seen, and how modeling the change fixes it.
- When you'd pick Huber or quantile loss over squared error.

6 Evaluation

6.1 Walk-forward (expanding-window) backtests

```
Test 2022:  train on 2017-2021           -> predict 2022
Test 2023:  train on 2017-2022           -> predict 2023
Test 2024:  train on 2017-2023           -> predict 2024
...
```

Random k-fold cross-validation shuffles days, so a model could train on July 15 and be tested on July 14. With autocorrelated market data that leaks the answer. Walk-forward respects time: every prediction uses only the past, mimicking real deployment. It also shows how performance changes by year, which is how the 2022 regime problem became visible.

6.2 Automated leakage tests

`tests/test_pipeline.py` builds fake data where each day's demand equals its date number, so you can read off exactly which day a feature came from. It checks two things:

- Every lagged fundamental comes from a day **strictly before** the trade date.
- The backtest never trains on data from the year it's testing (a 'spy' model records the latest training year it saw).

Testing for leakage with synthetic data, instead of trusting the code, is a habit worth copying in any forecasting project.

6.3 Point-forecast metrics

Metric	Definition	Reads as
MAE	average of |prediction - actual|	Typical miss in $/MWh. Robust, easy to explain
RMSE	square root of average squared error	Like MAE but punishes big misses more. RMSE much larger than MAE means a few huge misses
Bias	average of (prediction - actual)	Systematic over- or under-prediction. The level-LightGBM had +9.95: it was stuck too high after 2022
Directional accuracy	% of days the model called the up/down move right	What a trader using the model to pick direction cares about

6.4 Interval coverage (calibration of bands)

If the P10-P90 band is honest, 80% of actual prices should fall inside it. Results were 67-75% at all hubs: the bands are **too narrow**, meaning the model is overconfident. Fixes include conformal prediction or widening bands using recent errors. You shouldn't size risk off these bands until they're calibrated.

6.5 Classification metrics (T03)

- **Base rate:** how often spikes happen at all (11%). Every result should be compared to it.
- **AUC (ROC area):** the probability that a randomly chosen spike day gets a higher score than a randomly chosen normal day. 0.5 is a coin flip, 1.0 perfect. It measures *ranking*, not whether the probabilities are correct.
- **Brier score:** average squared difference between predicted probability and outcome (0 or 1). Lower is better. Compared against always predicting the base rate.
- **Precision in the top 10%:** of the days the model is most worried about, how many were spikes. 66% vs an 11% base rate. This is the most trader-friendly number: 'when it warns you, it's right two times out of three'.

6.6 Choosing a fair baseline and a fair label

THE KEY IDEA

A good result needs a baseline that's genuinely hard to beat. For spike days, 'yesterday was stretched' alone scored AUC 0.84 because heat waves last several days. The model's real contribution is the gap: 0.93 vs 0.84.

Label definitions can quietly create fake results. The first spike label (heat rate above last year's 90th percentile) flagged 101 of 232 days in 2024. Cheap gas that year inflated *every* day's heat rate, so 'spike' just meant 'it's 2024'. The fix was a relative label: at least 1.5x the trailing 30-day median.

CHECK YOURSELF: CAN YOU EXPLAIN

- Why walk-forward and not random cross-validation.
- The difference between MAE and RMSE, and what a large bias tells you.
- What 72% coverage on an 80% band means.
- What AUC measures and what it doesn't.
- Why a 'yesterday-only' baseline was necessary for T03.

7 The theses, walked through

T01: Solar cannibalization at SP15

Question	Does more CAISO solar push the SP15 on-peak heat rate down?
Method	Yearly and monthly heat-rate tables; OLS of IHR on solar, wind, hydro, demand with month and weekday fixed effects and HAC s.e.; solar x year interaction; LightGBM partial dependence
Result	Spring median IHR 10.3 (2019) to 2.1 (2026) while spring solar share rose from 15% to 25%. Pooled: +10 points of solar is -7.85 (CI -9.39 to -6.31), n = 1,366, R^2 = 0.41
Limits	Henry Hub denominator distorts 2022-23; the on-peak block blends midday and evening, so true midday cannibalization is larger

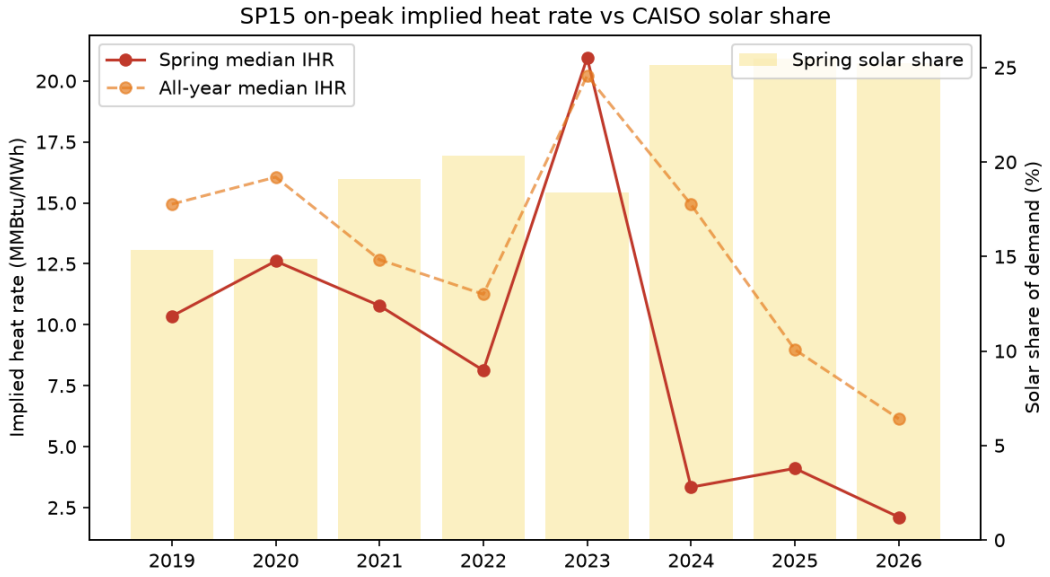


Figure 1. *T01: spring heat rate falls as solar share rises. The 2023 spike is the western gas crisis, not a reversal.*

T01b: Do batteries blunt cannibalization?

Question	Is solar's per-unit price effect shrinking because batteries absorb the midday surplus?
Method	Recovered CAISO batteries from hourly 'Other' data; fleet-capability proxy; solar x battery interaction; robustness against a time trend and without 2022-23; delta-method marginal-effect chart
Result	Not supported. Interaction -0.012 (p = 0.59); +0.153 with a time trend (p = 0.11); -0.0004 excluding 2022-23 (p = 0.98)
Why it matters	Batteries shift energy from noon to evening, both inside the HE7-22 block, so the block average barely moves. Their effect lives in intraday shape, which needs hourly prices. Part 8 (thesis E2) runs exactly that test with ERCOT's hourly prices. The apparent 'flattening' in T01 was mostly the 2022-23 gas distortion.

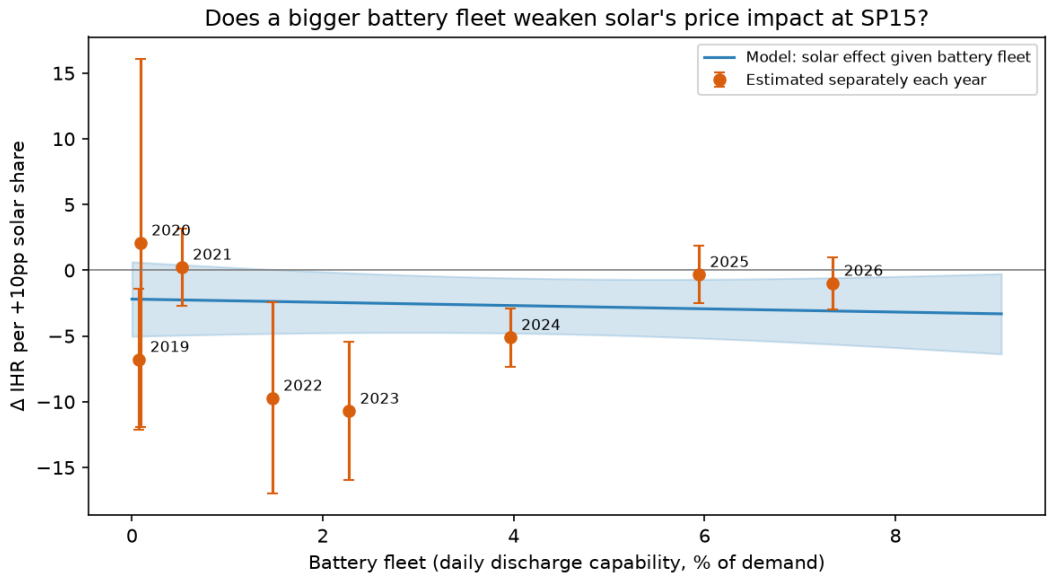


Figure 2. *T01b: if batteries explained the per-year slopes (orange), they would follow the blue line. They scatter around it instead.*

T02: The Mid-C vs SP15 spread is a hydro story

Question	Does Northwest hydro drive the Mid-C minus SP15 spread, and is it usable on the trade date?
Method	Hydro anomaly vs monthly climatology; OLS with BPA wind and load, CAISO solar and load, gas, fixed effects, HAC; quintile table; walk-forward ridge on the spread change using yesterday's anomaly
Result	+1 GW above normal: -$5.99/MWh (CI -6.82 to -5.17), $R^2 = 0.55$. Very dry days +$16.8, very wet days -$8.1. Forecast improvement over persistence: 4%
Surprise	Mid-C is above SP15 in spring on average, because solar crushes SP15 harder. Season alone is the wrong signal; the anomaly is the right one. Mid-C flipped from -$4.6 (2019-22) to +$15.3 (2023+).
Limits	Short, drought-heavy history for 'normal'; intertie transmission limits aren't in EIA data; regional gas differs from Henry Hub

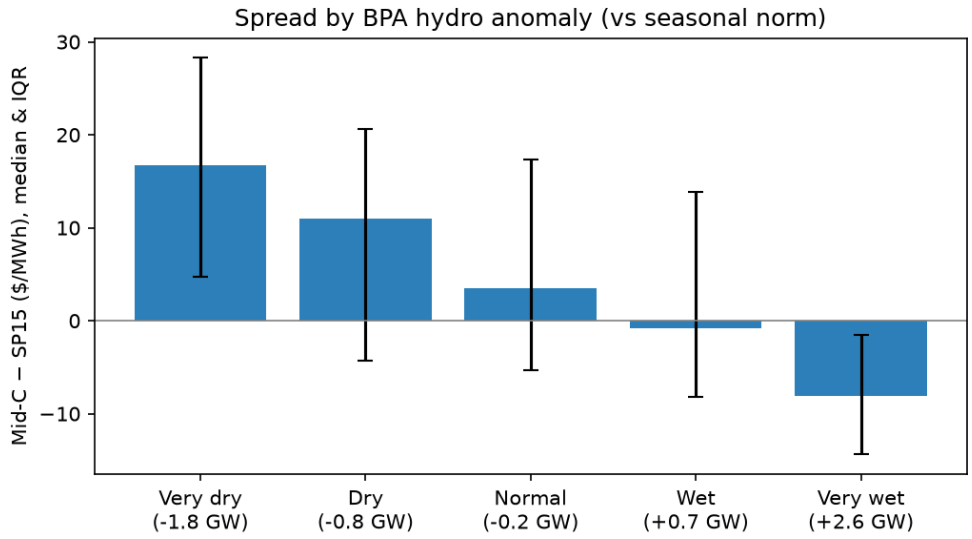


Figure 3. *T02: the spread falls steadily from very dry to very wet days.*

T03: PJM West convexity and spike days

Question	Is the heat rate convex in load, and can spike days be seen coming?
Method	Summer quantile regression (P10/P50/P90) on load and load²; load-decile table; 'excess dollars' concentration; walk-forward logistic and LightGBM classifiers with a yesterday-only baseline
Result	P90 slope rises from 1.02 to 1.53 per GW between 94 and 121 GW. Top load decile = 39% of summer upside. AUC 0.93 (logistic), 0.91 (LightGBM), 0.84 baseline; 66% precision in the top 10% vs 11% base rate
Limits	This hub's price is the real-time index; RTO-wide load misses West-zone congestion; no implied-volatility data, so 'underpriced' can't be tested

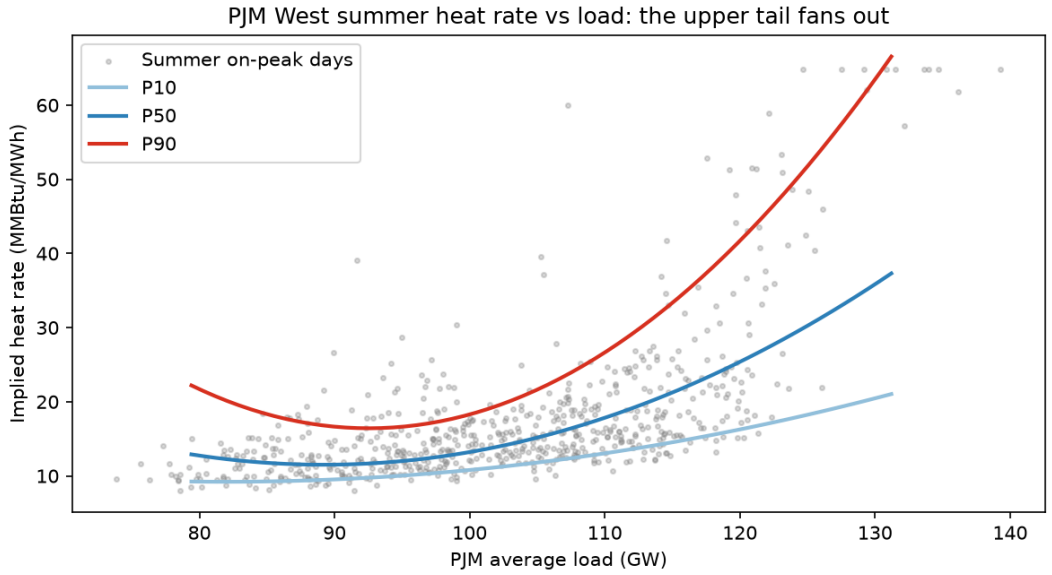


Figure 4. *T03: the P90 line pulls away from the median as load rises. That is convexity.*

F01: Next-day forecast benchmark

Hub	Persistence MAE	Best model	Best MAE	Improvement
PJM West	15.12	Ridge on heat rate	12.07	20%
Mass Hub	15.16	Ridge on heat rate	13.12	13%
SP15	11.04	LightGBM on change	10.12	8%
Palo Verde	15.20	LightGBM on change	14.75	3%
Mid-C	20.21	LightGBM on change	19.71	2%

Interpretation (a hypothesis, not yet tested): eastern prices follow gas and load, which the models see. Western prices depend on hydro, solar and regional gas prices that Henry Hub misses, so fundamentals add little there.

8 ERCOT: working with hourly data

ERCOT runs the Texas grid, the fastest-changing power market in the U.S.: solar and batteries are being added faster there than anywhere else. This part extends the project to ERCOT. It needed a new data source, and because that source is **hourly**, it opens up questions the daily EIA prices can't answer.

8.1 A second data source

EIA's price files stopped carrying ERCOT after 2018. ERCOT publishes its own market data, and one public report, *Historical DAM Load Zone and Hub Prices* (report NP4-180-ER), has every hourly day-ahead price at every trading hub back to 2010. No account or key is needed.

- **Finding the files:** the report page lists one zip per year, each with a numeric document ID. `ercot._yearly_doc_ids()` reads that page and pulls out the IDs with a regular expression, so the code finds new years automatically instead of hard-coding links.
- **Inside each zip:** one Excel workbook with a sheet per month and one row per hour per settlement point. The project keeps the six trading hubs (North, Houston, South, West, Panhandle, and the hub average) and uses **HB_HUBAVG**, the average of the four main hubs, as ERCOT's headline price.
- **Caching:** each year is parsed once and saved as Parquet. Only the current year is re-downloaded, because it's still growing.
- **Grid data:** still EIA-930 (balancing authority ERCO), now pulled *hourly*: demand, solar, wind, gas generation, and the 'Other' and 'Battery' categories.

CAUTION

Data arrives messier than you expect. Two problems showed up here: in some years ERCOT's price column loaded as text, not numbers, and Henry Hub's gas column did the same. Both quietly broke the math until they were converted with `pd.to_numeric`. Check column types (`df.dtypes`) every time you load a new source.

8.2 Lining up hourly data

With daily data, matching two sources only means matching dates. With hourly data, you also have to agree on what an 'hour' is, and on what time zone it is in.

- **Hour ending:** ERCOT labels hours by when they *end*. 'HE20' is 7pm to 8pm. HE1 is midnight to 1am, and there is no HE0.
- **Time zones:** EIA-930's hourly data is in UTC, also labeled by the end of the hour. The code steps each EIA timestamp back one hour (to the start of the hour), converts UTC to US Central, and then the local date and the start hour + 1 give ERCOT's delivery date and hour ending.
- **Daylight saving time:** on the November fall-back day, 1am happens twice, so ERCOT lists that hour twice with a 'repeated hour' flag. The code drops the repeat so every day has at most 24 hours. Days with fewer than 23 matched hours are skipped.

THE KEY IDEA

Always run a **sanity check** after an alignment step. Here: Texas solar output should peak in the early afternoon. After conversion, average solar peaked at HE13 to HE15 every year, which is exactly right. If the time-zone logic had been off by an hour or more, the solar peak would have landed in the wrong place.

8.3 New ideas used in the ERCOT theses

Idea	What it means	Where it's used
Net-load ramp	How much net load climbs from its midday low (HE9-16) to its evening high (HE17-23), in GW. The steeper the ramp, the more flexible supply the grid needs right after sunset.	E2
Evening premium	Average HE19-22 price minus average HE10-15 price, divided by gas: how much more evening power costs than midday power, in heat-rate units.	E2
Excluding an extreme event	Winter Storm Uri (Feb 10-20, 2021) pinned prices at the $9,000 cap for days. That's a different physical regime, so those days are dropped, not just winsorized.	E1, E2
Log of spikes	Price spikes are multiplicative (a bad day can be 5x a normal one), so E1 models log(max price / gas). A 0.1 change in the log is about a 10% change.	E1
Standardized coefficients	scale(x) rescales a variable to mean 0, standard deviation 1. Two drivers in different units (or two very similar ones) can then be compared head to head: the bigger coefficient matters more.	E1
Comparing R^2	Fit two one-driver models on the same data. The driver whose model explains more of the variation is the better single explanation.	E1
Testing a difference	To ask whether driver A gained on driver B, test whether coefA - coefB = 0 (statsmodels t_test). Two separate p-values can't answer that.	E1
A seam in the data	When a source changes how it reports something (EIA started a separate battery series in Nov 2024), build the measure from the part that's visible on both sides, and check the chart for a jump at the seam.	E2

8.4 E1: ERCOT scarcity moved from the demand peak to the net-load peak

Question	Is the most expensive hour of the day now set by net load (after sunset) rather than by demand (late afternoon)?
Method	Average price and net-load shapes by hour and year; the share of days whose price peak falls within an hour of each peak; a per-year 'horse race' with both peaks in one model (standardized); a pooled test of whether net load gained on demand over time
Result	The price peak moved from HE17 to HE20 in 2023. Days priced within an hour of the demand peak fell from 74% (2019) to 13% (2026), while 85-89% now sit near the net-load peak. Net load's R^2 advantage over demand grew from +0.15 (2019-21) to +0.26 (2023+). Gap test: +0.087 per year, $p = 0.004$.
Twist	Net load's own effect didn't grow (+0.015 per year, $p = 0.37$). It was already the stronger driver in 2019, because wind made net load differ from demand even before solar. What changed is that demand alone **lost** its explanatory power (-0.072 per year, $p < 0.001$).
Limits	Day-ahead prices only; real-time scarcity is spikier. EIA net load excludes rooftop solar.

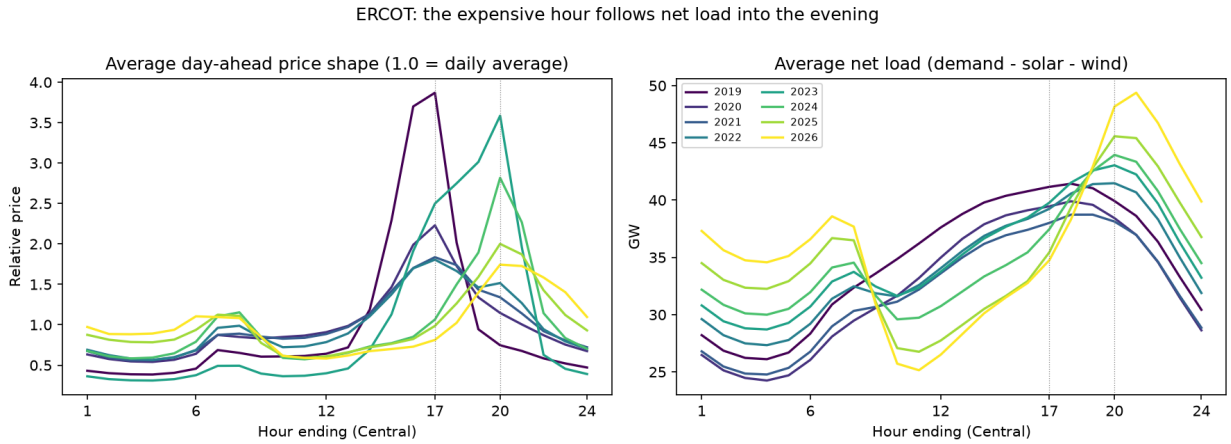


Figure 5. *E1: in 2019 the price spike sat at HE17 with the demand peak. By 2026 it sits at HE20-21, where net load now peaks. The midday dip in net load is solar.*

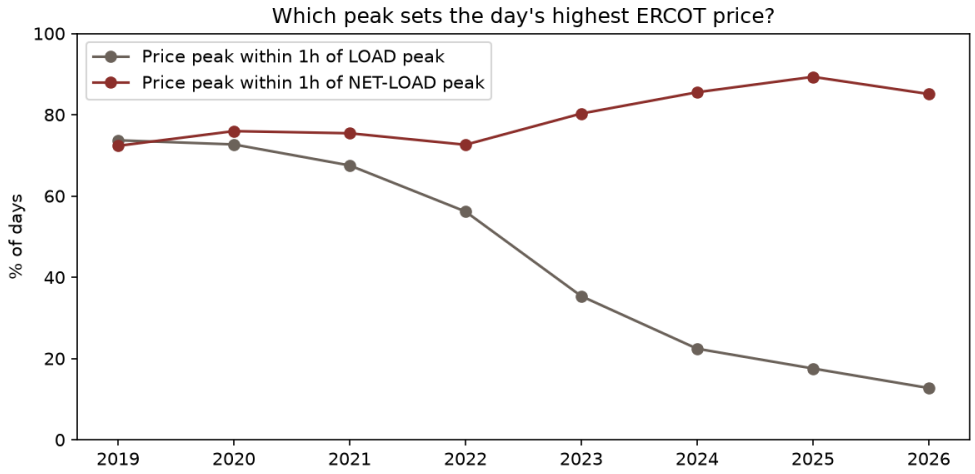


Figure 6. *E1: the share of days whose price peak lines up with the demand peak collapsed after 2022.*

8.5 E2: batteries are flattening the evening premium

This is the test T01b couldn't run. T01b concluded that batteries change the *shape* of the day, which a daily block price can't show. ERCOT's hourly prices make the shape visible, so the question can be asked directly.

Question	Does a bigger battery fleet make each GW of evening ramp cost less?
Method	Evening premium regressed on ramp, battery fleet, and ramp x battery, with controls for net-load peak, wind, month and weekday (HAC s.e.). Checked against a ramp x time trend, without 2022's gas spike, and on 2023+ only. Per-year ramp slopes plotted against the fleet, the same picture as T01b.
Battery measure	Same idea as CAISO: recover storage from EIA-930. Before Nov 2024 ERCOT batteries appear only as evening discharge inside 'Other'; after, as their own series. Evening discharge is visible in both, so the proxy uses that: trailing 90-day 95th percentile, lagged a day. Grows from about 0.03 GW (2019) to 8.7 GW (2026), with no jump at the seam.
Result	Ramp x battery is negative and significant in every version (main: -0.042, p = 0.003; 2023+: -0.112, p < 0.001). Per GW of ramp, the premium fell from +2.21 (2023) to +0.13 (2026) while the ramp itself grew from 15 to 27 GW.
Caution	Before 2023 the ramp didn't drive the evening premium at all, because scarcity still sat at 5pm (E1). So the evidence comes from four years of one build-out. Adding a time trend flips the plain ramp coefficient negative, a sign that battery growth and time are too intertwined to split cleanly.

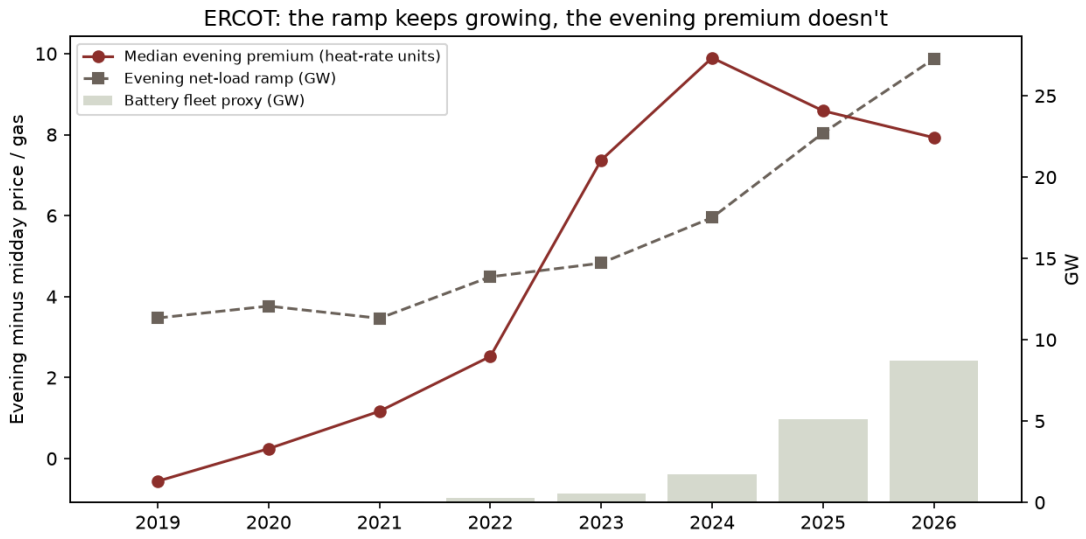


Figure 7. *E2: the evening ramp (dashed) keeps growing, but the premium (red) peaked in 2024 as the battery fleet (bars) took off.*

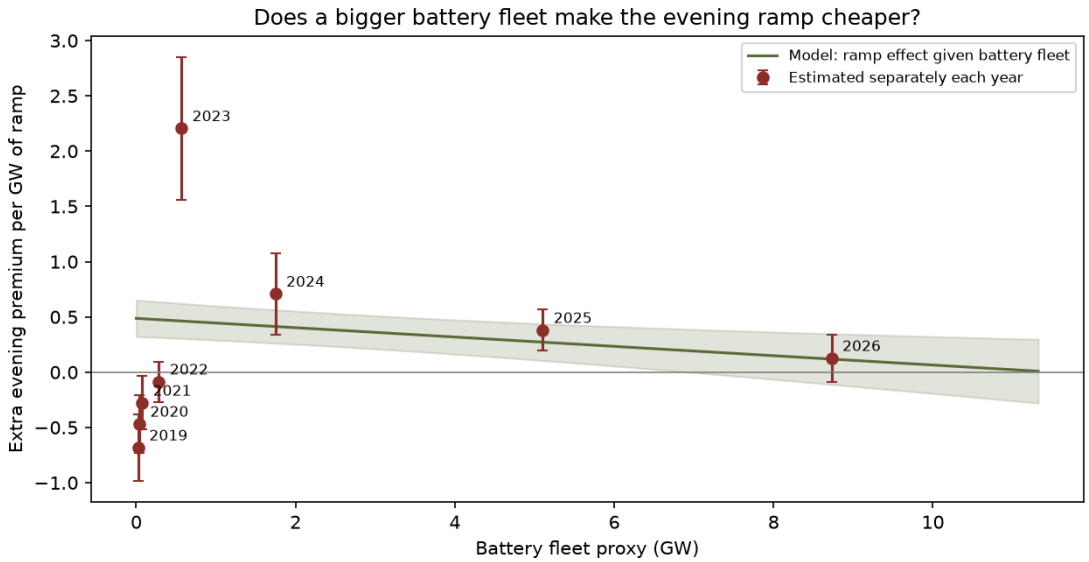


Figure 8. *E2: per-year cost of each GW of ramp. 2019-22 sit near zero (scarcity wasn't in the evening yet); from 2023 on, the cost falls as the fleet grows.*

8.6 How the pieces connect

- **T01 (CAISO):** solar pushes daily prices down relative to gas.
- **T01b (CAISO):** batteries don't show up in the daily price, probably because they reshuffle value *within* the day.
- **E1 (ERCOT):** with hourly data you can see where scarcity went: from the afternoon to the evening net-load peak.
- **E2 (ERCOT):** and batteries are now competing away that evening scarcity, the intraday effect T01b predicted but couldn't measure.

That chain is the point of a thesis-driven project: each result, including the negative one, sets up the next question.

CHECK YOURSELF: CAN YOU EXPLAIN

- Why ERCOT needed a different data source, and how the code finds ERCOT's yearly files.
- What 'hour ending' means, and the three steps that line EIA's UTC hours up with ERCOT's local hours.
- How the solar-peak check catches a time-zone bug.
- Why E1 tests the *difference* between two coefficients instead of each one alone.
- How the battery measure bridges EIA's change in reporting.
- Why E2's evidence really comes from 2023 onward, and what the flipped ramp coefficient warns you about.

9 Beyond EIA: working with the ISOs' own data

EIA is a convenient starting point, but practitioners work from the grid operators' own data: hourly and 5-minute prices, generation by fuel, reserve prices, and real-time vs day-ahead spreads. This part moves the project onto that data, prompted by advice from a power trader, and adds three studies that only become possible with it.

9.1 The new data sources

Source	What it adds	How it's fetched	Code
ERCOT real-time prices (NP6-785-ER)	15-minute real-time prices at every hub, so day-ahead can be compared with what actually cleared	ERCOT's public yearly files, averaged to hourly	ercot.rtm_hub_prices()
ERCOT ancillary prices (NP4-181-ER)	Hourly prices for each reserve product: REGUP, REGDN, RRS, ECRS, NSPIN	ERCOT's public yearly files	ercot.as_prices()
CAISO fuel mix and load	5-minute generation by fuel, with batteries reported directly (charging and discharging)	The open-source gridstatus library, no key	caiso.hourly_grid()
CAISO hourly prices (OASIS)	Hourly day-ahead prices at SP15, instead of EIA's one daily number	gridstatus; CAISO keeps only about 3 years (mid-2023 on)	caiso.dam_hub_prices()

- **One loader for many reports.** ERCOT's day-ahead, real-time and reserve archives share a layout (one zip per year, linked by a changing ID), so `ercot._yearly()` handles all three. Each report only needs a small parser. One 2026 workbook added seven title rows above the header, so the reader now looks for the 'Delivery Date' row instead of assuming row 1.
- **gridstatus.** An open-source Python package that hides each ISO's quirks: CAISO's zipped CSVs, 31-day request limits and rate limits. Its company also sells a hosted API with a longer price history, which is the way to get CAISO hourly prices before mid-2023.
- **Caching missing months.** When CAISO's archive has nothing for a month, the loader saves an empty file so later runs don't keep asking.

> THE KEY IDEA
>
> **How an error in EIA's data was caught.** Solar output peaks around noon, so the solar-weighted average hour of the day should stay put from year to year. For ERCOT it did (hour-ending 13.8 to 14.1 every year). For CAISO's EIA series it drifted from about 12.8 to 14.4 between mid-2022 and 2025, then snapped back. The sun didn't move; EIA's CAISO timestamps did. Switching to CAISO's own fuel mix removed the problem. This one-line check (where does solar peak?) is worth running on any hourly energy dataset.

9.2 D1: the duck curve, CAISO vs ERCOT

The **duck curve** is the daily shape of net load: a deep midday belly when solar peaks, then a steep evening neck when the sun sets. Batteries charge in the belly and discharge into the neck. So the load gas plants actually have to follow, the **residual load** (net load minus storage output), should have a flatter neck than net load itself.

Question	Is storage reshaping the duck in both markets, is CAISO further along, and do evening prices follow the storage-adjusted ramp?
Measures	Evening ramp = evening (HE17-23) peak minus midday (HE9-16) trough, for net load and for residual load. Storage shave = share of the net-load ramp storage removes. Belly depth = how far midday net load sinks below peak demand.
Result	Storage shave rose from 0% to 55% in CAISO and from 0% to 39% in ERCOT (2019 to 2026). CAISO's raw ramp grew from 13.9 to 26.9 GW, yet the ramp gas plants face shrank from 15.6 GW (2022) to 12.2 GW. CAISO's belly is now so deep that on many days midday solar and wind exceed demand.
Part 3	Not supported: the storage-adjusted ramp doesn't explain the evening premium better than the raw ramp. But the test is biased (see below).

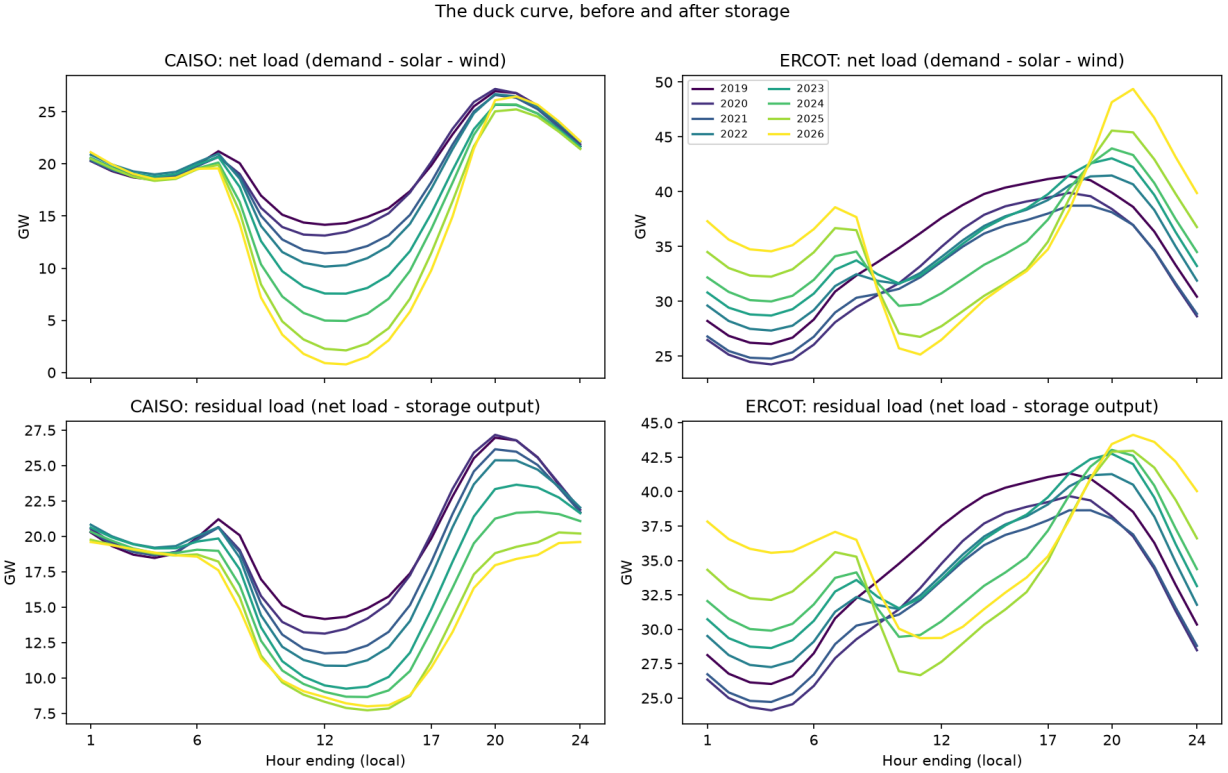


Figure 9. *D1: top row, the raw duck; bottom row, after storage. CAISO's evening neck has dropped from about 27 to about 18 GW.*

CAUTION

Why part 3 misleads. Residual load subtracts what batteries actually did that day, and batteries discharge more on days when evening prices are high. So on the most expensive evenings the residual ramp looks smaller, weakening its link to price. That's reverse causality, the same trap E2 avoided by measuring fleet *capability*. A test can fail because the idea is wrong or because the test is wrong; always check which.

9.3 E3: real time vs day-ahead in ERCOT

Every hour clears twice: day-ahead (DA) the day before, and real-time (RT) on the day. **DART** = RT price minus DA price. Virtual traders trade exactly this gap. A **dec** buys in DA and sells back in RT, so it wins when RT spikes above DA. An **inc** sells in DA and buys back in RT, so it wins when DA turns out too high.

Question	Is ERCOT's battery fleet shrinking evening real-time spikes above day-ahead?
Method	Daily evening upside DART: the average of max(RT - DA, 0) over HE19-22, divided by gas. Regressed on the battery fleet with the evening ramp, net-load peak, wind, month and weekday as controls (HAC). Checked with a time trend, without 2022, 2023 on only, and at the 90th percentile (quantile regression).
Result	Negative and significant in all five versions: each GW of batteries lowers evening upside DART by about 0.50 heat-rate units, roughly $1.79/MWh at 2026 gas.
The pattern	Evening DA usually cleared above RT (7 of 8 years), an insurance premium against rare RT spikes. That premium peaked at $17.49/MWh in 2023 and fell to $0.72 by 2026. Spikes got much smaller (standard deviation 325 to 54 $/MWh) but only somewhat rarer (7.0% to 4.5% of hours).

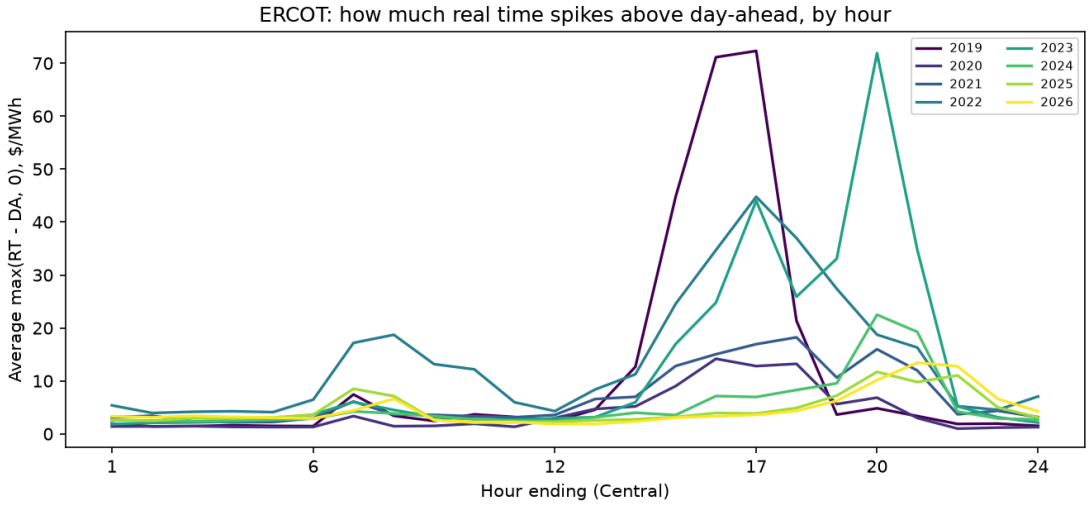


Figure 10. *E3: how far real time spikes above day-ahead, by hour. The evening hump peaked in 2023 and has collapsed since.*

9.4 E4: reserve prices in ERCOT

Ancillary services pay a resource to stand ready rather than to produce energy. Prices are $ per MW per hour of standby capacity. Batteries are ideal for the fast products (Regulation and Responsive Reserve), and the early Texas fleet earned most of its money there.

Question	Have batteries saturated ERCOT's reserve markets?
Method	Log of each product's daily price, regressed on the battery fleet with log gas, net-load peak, month and weekday as controls (HAC); checked with a time trend and 2023 on only. A log model reads as '% change per GW'.
Result	Prices fall about 26% (REGUP) and 30% (RRS) per GW of fleet, and still 19% with a time trend. RRS went from $17.11/MW-h (2021) to $0.71 (2026). Illustratively, selling RRS every hour was worth about $186/kW-year in 2023 and $14 in 2026, while daily arbitrage fell from about $120 to $33.
Caveat	ERCOT changed how much reserve it buys (more after Uri, ECRS added mid-2023, trimmed since). ECRS's own time-trend row even flips sign, because ECRS only exists in the battery build-out years.

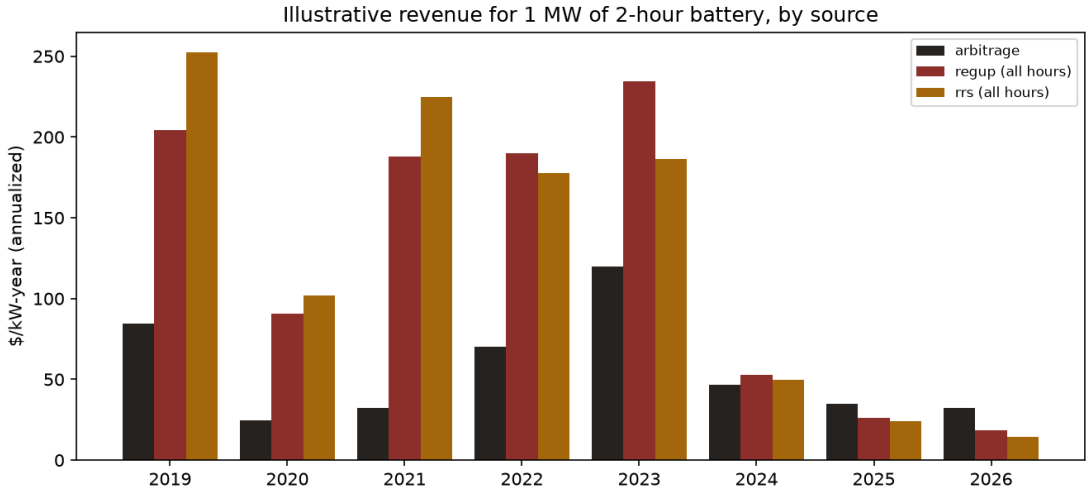


Figure 11. *E4: illustrative revenue for 1 MW of 2-hour battery. Reserve revenue collapsed after 2023; arbitrage is now the larger line, and it's shrinking too.*

9.5 How D1, E2, E3 and E4 fit together

- **D1:** storage is physically reshaping the evening, most in CAISO.
- **E2:** in ERCOT, that's compressing the day-ahead evening premium.
- **E3:** and it's removing the real-time spikes that made the evening risky.
- **E4:** meanwhile the reserve markets that early batteries relied on have saturated.

Together: **every revenue line a battery depends on is being competed away by other batteries.** That's the cannibalization theme that runs from T01 (solar) through to storage.

9.6 Physical vs financial power trading

Power trading splits into two worlds. The project touches both, but most of its tests are on the physical side.

	Physical trading	Financial trading
What it is	Managing the actual production and delivery of electricity	Taking positions on prices without delivering power
Core topics	The supply stack and dispatch by heat rate and fuel cost; scheduling and offering generation into day-ahead and real-time markets; ancillary services	Virtual bids (incs and decs) on the DA/RT spread; spread trades on congestion between two nodes (Up-To-Congestion, Point-to-Point); Financial Transmission Rights (FTRs, called CRRs in ERCOT and CAISO) that hedge or bet on congestion over months
Where it appears here	T03 (supply stack convexity), E1 and E2 (dispatch into the evening), E4 (ancillary services)	E3 (the DA/RT spread that incs and decs trade). Congestion trades (UTC, PTP, FTR/CRR) aren't modeled yet; they need node-level prices

CHECK YOURSELF: CAN YOU EXPLAIN

- Why ISO data beats EIA for this kind of work, and what each new source adds.
- How the solar-peak check exposed the timing error in EIA's CAISO data.
- What residual load and storage shave measure, and why part 3 of D1 is a biased test.
- What DART is, and who wins on each side: an inc or a dec.
- Why reserve prices are modeled in logs, and why ECRS's time-trend result can't be trusted.
- The difference between physical and financial trading, with one example of each from the project.

10 Research lessons: eight times the first answer was wrong

These are worth knowing cold. They show the difference between running models and doing research. Each is a mistake that is easy to make and easy to miss.

#	First result	What was wrong	Fix	Lesson
1	LightGBM lost to 'tomorrow = today' at SP15	Trees can't predict outside their training range; 2022 prices were unseen	Model the daily change, not the level	Check whether your model class can represent the regime you're testing on
2	'Solar's effect is flattening, probably batteries'	The steep 2022-23 slopes came from the gas-basis distortion, and batteries didn't explain the rest	Ran T01b properly; reported 'not supported'	A pattern in noisy yearly estimates isn't a trend. Test the explanation; report negatives
3	Spike-day AUC 0.88	Cheap 2024 gas made 44% of days 'spikes', and heat waves are persistent	Relative spike label; yesterday-only baseline	Inspect your label's base rate by year; always compare to the dumbest reasonable model
4	E1: 'net load's effect isn't growing' (p = 0.37)	The test asked the wrong question. The thesis is about net load gaining *relative to* demand	Test the difference of the two trend terms: +0.087/yr, p = 0.004	Make the statistical test match the claim. Then report what actually moved (demand lost relevance)
5	E2: one straight line across 2019-2026	Before 2023 the ramp didn't set evening prices at all, so pooling mixed two regimes	Per-year slopes, a 2023+ model, and an explicit caution in the verdict	Plot the per-period estimates before trusting one pooled number
6	CAISO's duck from EIA data looked fine at a glance	EIA's CAISO timestamps drifted up to 1.5 hours (2022-2025)	Checked where solar peaks each year; switched to CAISO's own data	Run a physical sanity check on every hourly dataset
7	D1 part 3: 'prices don't follow the storage-adjusted ramp'	Same-day battery dispatch reacts to prices (reverse causality)	Reported as not supported and biased; pointed to E2's capability-based test	A failed test can mean a flawed test. Say which
8	E3's pre-written story: decs profit from spikes that are getting rarer	Data showed evening DA usually above RT, and spikes shrank in size more than frequency	Rewrote the interpretation from the numbers	Write conclusions after the data, not before

The general workflow

- State the thesis as something that could turn out false, with a predicted sign.
- Pick the cleanest comparison (fixed effects, anomalies, within-year variation).
- Look at the data and the charts before trusting a number.
- Try to break it: robustness checks, competing explanations, excluding odd periods.

- Let the code write the verdict from the numbers, so the conclusion can't drift from the evidence.
- Write down the limits and the trade expression, labeled as a hypothesis.

11 Code map

File	What it does	Concepts
powerml/config.py	Paths, hub definitions, hub-to-BA map, fuel codes	Hubs, BAs
powerml/data.py	Download and cache ICE prices, Henry Hub, EIA-930 daily and hourly; recover CAISO batteries	APIs, pagination, backoff, caching, time zones
powerml/features.py	Build one panel per hub; implied heat rate, shares, net load, calendar, lags; EXPLAIN vs KNOWN lists	Feature engineering, merge_asof, look-ahead bias
powerml/models.py	Persistence, HeatRateRidge, GBM, OnPersistence; walk_forward() and Backtest.scores()	Ridge, LightGBM, residual modeling, walk-forward, metrics
scripts/fetch_data.py	Pull and cache every input	Pipeline orchestration
scripts/forecast_benchmark.py	Model contest for one hub; scores, coverage, chart, report	Evaluation, quantile bands
scripts/thesis_01_*.py	Solar cannibalization	OLS, fixed effects, HAC, interactions, partial dependence
scripts/thesis_01b_*.py	Batteries	Interactions, delta method, confounding, reverse causality
scripts/thesis_02_*.py	Mid-C hydro spread	Climatology, anomalies, spreads, walk-forward ridge
scripts/thesis_03_*.py	PJM convexity and spikes	Quantile regression, classification, AUC, Brier
powerml/ercot.py	ERCOT hourly hub prices from ERCOT's site; hourly EIA-930 for ERCO; battery proxy; daily ERCOT panel	Scraping a report index, hour-ending alignment, time zones, DST
scripts/thesis_e1_*.py	ERCOT peak-hour shift	Daily shapes, standardized coefficients, R^2 comparison, difference test
scripts/thesis_e2_*.py	ERCOT batteries and the evening premium	Interactions, delta method, regime change, collinearity
powerml/caiso.py	CAISO hourly prices (OASIS) and CAISO's own load and fuel mix, via gridstatus	Month-by-month caching, rolling archives, rate limits
data.hourly_grid()	Shared hourly EIA-930 builder for any balancing authority (used for ERCOT)	UTC to local hour-ending alignment
scripts/thesis_d1_*.py	Duck curve, CAISO vs ERCOT	Residual load, storage shave, R^2 comparison, reverse causality
scripts/thesis_e3_*.py	ERCOT real-time vs day-ahead	DART, quantile regression, robustness checks
scripts/thesis_e4_*.py	ERCOT reserve prices	Log-price regressions, illustrative revenue stack

File	What it does	Concepts
tests/test_pipeline.py	Two leakage guards on synthetic data	Testing for leakage

Running it

```
.venv\Scripts\python scripts\fetch_data.py
.venv\Scripts\python scripts\thesis_03_pjm_convexity.py
.venv\Scripts\python scripts\forecast_benchmark.py --hub PJMW
.venv\Scripts\python -m pytest -q
```

Each script writes a `report.md` and charts into `reports/`. A good way to learn: change one thing (the spike threshold, the test years, a feature) and predict what will happen before you rerun.

12 Review questions

Try to answer each question before reading the answer. They mix concepts from every part of the guide.

Q. Summarize the project in three sentences.

A It uses free EIA data to test power-market claims and to forecast next-day on-peak prices at five U.S. hubs. It models the implied heat rate (power over gas) to strip out gas price moves. Its strongest result: PJM West's heat rate is convex in load, and spike days are predictable a day ahead (AUC 0.93 against 0.84 for a persistence baseline).

Q. Why study heat rate instead of price?

A Gas sets the marginal price most hours, so raw power prices mostly track gas. Dividing it out isolates grid fundamentals, and heat rate is the unit traders use for spark-spread and heat-rate positions.

Q. How does the project avoid look-ahead bias?

A Separate 'explain' and 'known' feature sets, with known features using only information dated before the trade date. Walk-forward backtests by year. And unit tests on synthetic data that check both.

Q. LightGBM first lost to persistence. Why, and what fixed it?

A Trees can't extrapolate beyond their training range, and 2022 prices were unseen. Predicting the change from the last price makes persistence the floor; with that change it beat persistence every year.

Q. Which thesis failed, and what does the failure teach?

A T01b: batteries were expected to weaken solar's price impact. CAISO battery data was recovered from EIA's 'Other' category, a capability proxy avoided reverse causality, and the effect wasn't there. Batteries move energy within the on-peak block, so daily block prices can't show their effect; hourly prices could.

Q. What does an HAC standard error fix?

A Autocorrelated errors. Market conditions persist for days, so ordinary standard errors would be too small and overstate significance.

Q. How could the T03 result be traded?

A Because the payoff is convex in load, the value is in optionality: PJM West on-peak calls or heat-rate calls when the model's spike probability is high relative to implied volatility. Testing whether that is actually profitable needs implied-volatility data the repo doesn't have.

Q. What are the most valuable next extensions?

A Hourly prices from CAISO OASIS to test batteries on intraday shape; regional gas prices to fix the western heat-rate distortion; calibrated forecast bands; and the ISO's own load forecast for a live version.

Q. Why did the simple ridge model win in the East?

A PJM and New England prices are largely gas plus load, which is close to linear in heat-rate terms, so a regularized linear model captures most of it with less variance than trees.

Q. Why is ERCOT a better place than CAISO to test the battery idea?

A ERCOT publishes hourly prices, so the midday-to-evening shape is visible. CAISO's EIA prices are one daily on-peak block, and batteries mostly move energy within that block, so their effect averages out.

Q. In E1, why test the difference between two trend terms?

A The claim is that net load gained on demand as the price driver. Each trend term alone answers a different question. The difference was significant (+0.087 per year, $p = 0.004$), and it turned out to come from demand losing relevance rather than net load gaining.

Q. What's the difference between an inc and a dec, and what did E3 find about them?

A A dec buys day-ahead and sells back in real time, so it wins when real time spikes. An inc does the opposite. Evening day-ahead usually cleared above real time, a premium for spike protection that decs paid and incs collected. As batteries removed the spikes, that premium fell from \$17.49 to \$0.72/MWh.

Q. Why did D1's third test fail, and does that mean the idea is wrong?

A Not necessarily. Residual load uses same-day battery dispatch, and batteries discharge more when prices are high, so the measure is partly caused by the price it's meant to explain. E2's capability-based measure is the clean test, and it supports the idea.

Q. What's the difference between AUC and precision in the top 10%?

A AUC measures ranking across all thresholds. Precision in the top 10% is what you'd get acting only on the model's strongest warnings: 66% against an 11% base rate.

13 Glossary

Term	Meaning
Ancillary services	Payments for standing ready (regulation, reserves) rather than producing energy
AUC	Probability a random positive case is ranked above a random negative one
Balancing authority (BA)	Entity that balances supply and demand in an area; EIA-930 reports by BA
Basis	Price difference between two locations
Brier score	Mean squared error of predicted probabilities
Climatology	The normal value for a time of year
Collinearity	Two regressors moving almost together, so a model can't split credit between them
Convexity	Effect that grows faster than linearly; here, price vs load
DART	Real-time price minus day-ahead price for the same hour
Day-ahead / real-time	Next-day market vs the price that clears in the moment
Dec / Inc	Virtual trades: a dec buys day-ahead and sells real time; an inc sells day-ahead and buys real time
Delta method	Way to get the standard error of a combination of coefficients
Duck curve	The daily net-load shape: deep midday belly from solar, steep evening neck
ECRS	ERCOT Contingency Reserve Service, a reserve product added in June 2023
Evening premium	Evening (HE19-22) price minus midday (HE10-15) price, divided by gas
Fixed effects	Separate intercepts per group (month, weekday) to absorb what's typical for that group
FTR / CRR	Rights that pay the congestion price difference between two nodes over months
HAC / Newey-West	Standard errors robust to autocorrelation and changing variance
Heat rate (implied)	Power price / gas price, in MMBtu/MWh
Hour ending (HE)	Hours labeled by when they end: HE20 is 7pm-8pm
Hub	Named trading location that stands in for a region's price
Huber loss	Squared error for small misses, absolute error for big ones
Leakage / look-ahead bias	Using information a real forecaster wouldn't have had
MAE / RMSE	Average absolute error / root mean squared error
Merit order / supply stack	Plants ordered by marginal cost; the last one needed sets the price
Net load	Demand minus wind and solar
Net-load ramp	Rise in net load from its midday low to its evening high, in GW
On-peak	Weekday hour-ending 7-22 block

Term	Meaning
Partial dependence	A model's average prediction as one feature varies
Persistence	Forecast that tomorrow equals today
Pinball loss	Asymmetric loss used to fit quantiles
Quantile regression	Regression for a chosen percentile rather than the mean
Regularization	Penalty on model complexity to reduce overfitting
Residual load	Net load minus storage output: what other plants must follow
RRS / REGUP	Responsive Reserve and Regulation Up, ERCOT's fast reserve products
Spark spread	Power price minus (plant heat rate x gas price)
Standardized coefficient	Effect of a 1-standard-deviation change, so different drivers can be compared
Storage shave	Share of the evening net-load ramp removed by storage
Walk-forward	Train on the past, test on the next period, roll forward
Winsorize	Cap extreme values at a chosen percentile
Winter Storm Uri	Feb 2021 Texas freeze; prices sat at the $9,000 cap for days

Study plan: what to learn next

Topic	Why	Where to start
Merit order and LMPs	Foundation for everything	ISO market primers: PJM Learning Center, CAISO and ERCOT training pages
Regression inference	HAC, fixed effects, interactions	'Introductory Econometrics' (Wooldridge), chapters on OLS, inference and time series
Time-series forecasting	Baselines, backtesting, calibration	'Forecasting: Principles and Practice' (Hyndman, free online)
Gradient boosting	How LightGBM really works	LightGBM docs, 'Parameters Tuning' page; XGBoost paper intro
Quantile and probabilistic forecasting	Fixing the 72% coverage	Conformal prediction tutorials; pinball loss
Options on power	The T03 trade expression	Hull, 'Options, Futures, and Other Derivatives': options basics and volatility
CAISO market design	The 'why' behind CAISO's data: scheduling, settlement	CAISO's free training: caiso.com/stakeholder/training
Physical trading	Supply stack, dispatch, ancillary services	ISO training pages; search each term for practitioner walkthroughs
Financial trading	Virtuals (incs/decs), UTC and PTP spreads, FTRs/CRRs	ISO market manuals; congestion needs node-level prices, a natural next project
Multi-ISO data	Clean data across markets without each portal's quirks	gridstatus (open source) and gridstatus.io (hosted API)

Suggested exercises, in order: (1) recompute one number from a report by hand in a notebook; (2) rerun T01 excluding 2022-23 and see what changes; (3) recalibrate the P10-P90 bands so coverage reaches 80%; (4) rerun E2 on ERCOT real-time prices instead of day-ahead; (5) rerun E2 for CAISO using caiso.dam_hub_prices() (2023 on); (6) test whether ERCOT congestion between HB_WEST and HB_HOUSTON tracks West Texas wind and solar, the simplest congestion trade.